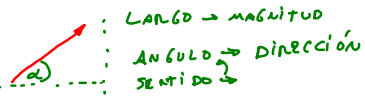
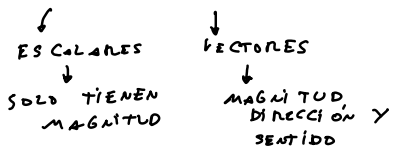


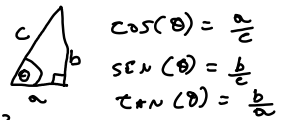
CANTIDADES FISICAS → FUERZA



SUMAR VECTORES GRÁFICA:

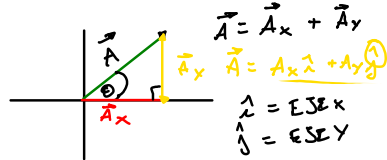


TRIANGULO RECTANGULO Y SU TRIGONOMETRIA



$$a^2 + b^2 = c^2$$

VECTORES EN EL PLANO CARTESIANO ANALITICA



$$\vec{A} = \vec{A}_x + \vec{A}_y$$

$$\vec{A} = A_x \hat{i} + A_y \hat{j}$$

$$\hat{i} = \text{ESE } x$$

$$\hat{j} = \text{ESE } y$$

COMPONENTES DE UN VECTOR

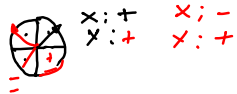
$$|\vec{A}| = \sqrt{A_x^2 + A_y^2}$$

$$A_x \hat{i} = |\vec{A}| \cos \theta$$

$$A_y \hat{j} = |\vec{A}| \sin \theta$$

$$\theta = \tan^{-1} \left(\frac{|A_y|}{|A_x|} \right)$$

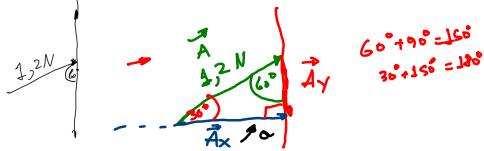
CIRCULO GONIOMETRICO



$$x: + \quad x: -$$

$$y: + \quad y: -$$

Una persona dispara un proyectil al cual choca con una pared, formando un ángulo de 30° con la horizontal. Calcule la magnitud del proyectil en 1.2 N, perpendicular hacia arriba a la derecha. (100)



$$\cos(30) = \frac{\text{cateto adyacente}}{\text{hipotenusa}}$$

$$\cos(30) = \frac{\text{cateto adyacente}}{1.2 \text{ N}} = \frac{A_x}{1.2 \text{ N}}$$

$$\vec{A}_x = \cos(30) \cdot 1.2 \text{ N} = \frac{3\sqrt{3}}{5} \text{ N}$$

$$\vec{A}_y = a^2 + b^2 = c^2$$

$$\left(\frac{3\sqrt{3}}{5} \text{ N} \right)^2 + b^2 = 1.2^2$$

$$b^2 = 1.2^2 - \left(\frac{3\sqrt{3}}{5} \right)^2$$

$$b = \frac{3}{5}$$

$$\vec{A} = \frac{3\sqrt{3}}{5} \hat{i} + \frac{3}{5} \hat{j}$$